

Analysis plots for the joint disturbance / process-error filter (ORJointUpdate in OR_joint_update2.Rmd)

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This note explains the diagnostic plots that `OR_joint_update_analysis.Rmd` makes from runs of `ORJointUpdate`, and gives the formula for every quantity they show:

1. filter divergence at the final observation of each site, with an alternative statistic (the predictive PIT) aimed at the same question;
2. the probability of detecting a disturbance in 2004 against the observed biomass loss from 2003 to 2004.

Section 1 defines the filter’s notation and Section 2 the one-step-ahead predictive. Sections 3 and 4 define the plotted quantities. Section 5 describes the experiment grid and the 5-site test run in `experiments/testRun`, and shows its plots.

1 The filter

1.1 Data, time axis and state

- **Sites and years.** s indexes sites (pixels of tile h03v04). The filter runs on annual timesteps $t = 1, \dots, T$, $T = 28$, where timestep t is calendar year $1989 + t$ (1990–2017; `atime`). The biomass product covers 1990–2017 and the disturbance product covers 1985–2012. Both are looked up by calendar year, so the product’s flag for year Y is applied at the timestep for year Y .
- **State.** $x_t = (x_{t,1}, x_{t,2}, x_{t,3})^\top$ is the vector of three carbon pools (kg C m^{-2}) of VSEM, the Very Simple Ecosystem Model in the R package `BayesianTools`. The code treats them as leaf, soil and wood. The wood pool $x_{t,3}$ is the one compared with above-ground biomass. One VSEM year of the state x , driven by that site’s daily photosynthetically active radiation (PAR, from Daymet), is written $f_t(x)$.
- **Observation.** y_t is the Landsat-based above-ground biomass (AGB) of the site in year $1989 + t$. It is converted from Mg ha^{-1} to kg C m^{-2} by $\times 0.5$ (biomass to carbon) $\times 0.1$. The observation model is

$$y_t \mid x_t \sim \mathcal{N}(Hx_t, R_t), \quad H = (0, 0, 1), \quad R_t = m_R y_t + b_R,$$

where R_t is the observation-error variance and (m_R, b_R) is `Rparams`. “Heteroskedastic” (`het`) runs have $m_R > 0$, so R_t grows with biomass. “Constant” (`con`) runs have $m_R = 0$.

1.2 Disturbance labels and their prior

Each year the site carries one label $k \in \{0, 1, \dots, D\}$:

- $k = 0$ means undisturbed.
- $k \geq 1$ is a disturbance sub-class. Each disturbance type of the product (other, cut, fire, biotic) is split into the clusters of a Gaussian mixture fitted to that type's fraction of wood retained after disturbance. There are $4 + 6 + 3 + 3$ clusters, so $D = 16$.
- Under label $k \geq 1$, the disturbance step d_k (`disturbance.p`) multiplies leaf and wood by random retention fractions. These are drawn from $\mathcal{N}((0.2, r_k), \text{diag}(0.04, v_k))$, where r_k and v_k are the cluster's mean and variance, truncated to $[0, 1]$. Of the carbon removed, 5% goes to the soil pool.

The prior probability of label k in year t is $\beta_{k,t}$ (`dist$p[k+1, t]`). It is built as follows:

- **Background.** Each type gets its background rate (`d.prior`). That rate is split across the type's clusters in proportion to the cluster weights.
- **With the disturbance product.** In a year the product flags with a type, that type's probability is replaced by the product's *user's accuracy* for the type (Kennedy et al. 2015): the fraction of the product's flags of that type that are correct (0.77 other, 0.92 cut, 0.78 fire, 0.78 biotic). It is then split the same way.
- **Undisturbed.** $\beta_{0,t} = 1 - \sum_{k \geq 1} \beta_{k,t}$.

1.3 Process error and the precision grid

Process error is added in the undisturbed label only. Its covariance is

$$Q(\tau) = \text{diag}(q_1^2, q_2^2, \tau^{-1}),$$

where q_1, q_2 are fixed leaf and soil standard deviations from the calibration (`fit.RData`). τ is the unknown *precision* (inverse variance) of the wood process error.

The filter places τ on a grid $\tau_1 < \dots < \tau_G$ of $G = 200$ points, spaced evenly in $\log \tau$ from 0.1 to 10^4 . The prior is $\text{Gamma}(\text{shape } 0.1, \text{rate } 2 \times 10^{-4})$. $\pi_0(g)$ is the prior mass of the grid cell around τ_g , with cell edges at the midpoints between neighbouring grid points, renormalised over the grid. $\pi_t(g)$ denotes the posterior probability of τ_g given $y_{1:t}$, the observations up to year t .

1.4 Posterior representation

Conditional on τ_g , the posterior of x_t is a mixture of two Gaussians, $j \in \{1, 2\}$, with:

- weights $w_t(g, j)$, where $w_t(g, 1) + w_t(g, 2) = 1$;
- means $m_t(g, j)$ and covariances $P_t(g, j)$.

Component $j = 1$ collects the undisturbed label and $j = 2$ the disturbance labels (see the projection below). At $t = 0$ every component is $\mathcal{N}(\bar{x}_0, P_0)$ with weight $\frac{1}{2}$, where $\bar{x}_0, P_0$ are the mean and covariance of an initial-condition ensemble. That ensemble pairs every site's 1990 AGB with leaf and soil pools from a VSEM regrowth run of matching wood biomass. It is therefore a cross-site distribution, not specific to the site being filtered.

1.5 Forecast step

1. **Proposal.** Draw $N = 100$ samples $\xi^{(1)}, \dots, \xi^{(N)}$ from the equal-weight mixture of all $2G$ components, rejecting samples with a non-positive pool. The proposal density is $q(\xi) = \frac{1}{2G} \sum_{g,j} \mathcal{N}(\xi; m_{t-1}(g, j), P_{t-1}(g, j))$.
2. **Propagation.** Push each sample through one VSEM year: $f_t(\xi^{(n)})$. For label $k \geq 1$, also apply the disturbance step, giving $d_k(f_t(\xi^{(n)}))$. For $k = 0$ the result is left unchanged. Write $z_k^{(n)}$ for the result.
3. **Importance weights.** Component (g, j) reweights the shared samples by

$$W_{gj}(n) \propto \frac{\mathcal{N}(\xi^{(n)}; m_{t-1}(g, j), P_{t-1}(g, j))}{q(\xi^{(n)})}, \quad \sum_n W_{gj}(n) = 1.$$

4. **Forecast moments.** The forecast moments of component (g, j) under label k are

$$\mu_{gjk}^F = \sum_n W_{gj}(n) z_k^{(n)}, \quad \Sigma_{gjk}^F = \frac{\sum_n W_{gj}(n) (z_k^{(n)} - \mu_{gjk}^F)(z_k^{(n)} - \mu_{gjk}^F)^\top}{1 - \sum_n W_{gj}(n)^2} + \mathbf{1}\{k = 0\} Q(\tau_g).$$

The denominator is the usual bias correction for a weighted sample covariance. We write

$$C_{gjk} = [\Sigma_{gjk}^F - \mathbf{1}\{k = 0\} Q(\tau_g)]_{33}$$

for the ensemble (sample) variance of the wood pool. The wood forecast variance is then $\Sigma_{gjk,33}^F = C_{gjk} + \mathbf{1}\{k = 0\} \tau_g^{-1}$.

5. **Degenerate components.** If $1 - \sum_n W_{gj}(n)^2 < 10^{-8}$, component (g, j) 's weights sit on essentially one sample, and its covariance cannot be estimated. This happens to components that are far from every proposal sample. Such a component is given zero likelihood in the analysis, so zero weight. Its moments are set to the unweighted sample moments as a finite placeholder. The forecast weight removed is stored per step as `hist$wDrop`. In the test run it occurred once (site 212, 2005, with no product and GEDI heteroskedastic R). The weight removed there was 10^{-119} .

1.6 Analysis step

For every (g, j, k) , a Kalman update with observation y_t gives the analysis moments and the component likelihood

$$\ell_{gjk} = \mathcal{N}(y_t; \mu_{gjk,3}^F, \Sigma_{gjk,33}^F + R_t).$$

The marginal likelihood of y_t given τ_g , the τ posterior, and the analysis weight of each (j, k) given τ_g are

$$Z_t(g) = \sum_{j,k} w_{t-1}(g, j) \beta_{k,t} \ell_{gjk}, \quad \pi_t(g) \propto \pi_{t-1}(g) Z_t(g), \quad w_t^A(g, j, k) = \frac{w_{t-1}(g, j) \beta_{k,t} \ell_{gjk}}{Z_t(g)}.$$

Projection. For each g , the $2(D + 1)$ analysis components are moment-matched down to two Gaussians:

- $j = 1$ from all components with $k = 0$, with weight $w_t(g, 1) = \sum_j w_t^A(g, j, 0)$;
- $j = 2$ from all components with $k \geq 1$, with weight $w_t(g, 2) = 1 - w_t(g, 1)$.

In a year without an observation, the analysis equals the forecast and $\ell_{gjk} = 1$.

Analysis summaries. The filter stores, for each year, summaries of the marginal posterior of the wood pool given $y_{1:t}$. That posterior is the Gaussian mixture

$$p(x_{t,3} | y_{1:t}) = \sum_{g,j,k} \pi_t(g) w_t^A(g, j, k) \mathcal{N}(x_{t,3}; \mu_{gjk,3}^A, \Sigma_{gjk,33}^A), \quad (1)$$

where μ_{gjk}^A and Σ_{gjk}^A are the Kalman analysis mean and covariance of component (g, j, k) . The stored summaries are:

- the analysis mean $\hat{x}_t = \mathbb{E}[x_{t,3} | y_{1:t}]$ (`hist$mean`);
- the analysis variance (`hist$var`);
- the 95% analysis interval $[L_t, U_t]$ (`hist$ci95`). L_t and U_t are the 2.5% and 97.5% quantiles of the mixture (1), found by root-finding on its cumulative distribution function. Because the mixture can be skewed (for example in years with a high disturbance prior), the interval need not be symmetric about $\hat{x}_t$.

2 The one-step-ahead predictive

The second statistic in Plot 1 uses the distribution of y_t given $y_{1:t-1}$, i.e. *before* y_t is assimilated. Under the filter above, this predictive is the mixture

$$p(y_t | y_{1:t-1}) = \sum_{g=1}^G \sum_{j=1}^2 \sum_{k=0}^D \omega_{gjk} \mathcal{N}(y_t; \mu_{gjk,3}^F, \Sigma_{gjk,33}^F + R_t), \quad \omega_{gjk} = \pi_{t-1}(g) w_{t-1}(g, j) \beta_{k,t}. \quad (2)$$

- **Integration over process error.** The sum over g uses π_{t-1} , the posterior of the process precision from the previous step. So the predictive averages over the process-error variance $1/\tau$ instead of fixing it.
- **Disturbance labels.** The sum over k uses the prior $\beta_{k,t}$, because the year- t label is not known before y_t is seen.
- **Weights.** The weights ω_{gjk} sum to one. Degenerate components get $\omega = 0$, and the rest are renormalised.

3 Plot 1: filter divergence and predictive PIT

Both statistics ask whether the filter still tracks the data at the end of the series, and both scale the error by the filter's own uncertainty. Let T_s be the last year with an observation at site s (2017 for every site here).

3.1 Filter divergence

Formula. The final observation minus the final analysis mean, divided by half the width of the 95% analysis interval:

$$\delta_s = \frac{y_{T_s} - \hat{x}_{T_s}}{(U_{T_s} - L_{T_s})/2}.$$

If the interval is symmetric about $\hat{x}_{T_s}$, $\delta_s = \pm 1$ exactly at its edges, and $|\delta_s| \leq 1$ means the final observation falls inside it.

How to read it. A calibrated filter should give $|\delta_s| \leq 1$ at most sites. A filter that has *diverged* has a confident analysis that sits away from the data, giving $|\delta_s| > 1$. The sign shows the direction: $\delta_s > 0$ means the analysis is below the observation. Two caveats:

- $\hat{x}_{T_s}$ has already assimilated y_{T_s} , so it is pulled toward the observation. That pull shrinks δ_s , so the statistic flags only strong divergence.
- $[L_t, U_t]$ is an interval for the latent wood pool. It does not include the observation error R_t , which acts in the other direction.

So $[-1, 1]$ is a rule of thumb, not an exact coverage level.

Panels. Page 1 of `exps_summary.pdf` plots δ_s against y_{T_s} , one point per site and one panel per experiment. The band $[-1, 1]$ is shaded, and the panel title gives the share of sites inside it.

3.2 Predictive PIT (alternative statistic)

Formula. The *probability integral transform* (PIT) of the observation y_t is the predictive cumulative probability of the value actually observed:

$$u_t = \Pr(y'_t \leq y_t \mid y_{1:t-1}) = \sum_{gjk} \omega_{gjk} \Phi\left(\frac{y_t - \mu_{gjk,3}^F}{\sqrt{\Sigma_{gjk,33}^F + R_t}}\right) \quad (\text{hist\$pit}),$$

where y'_t is a draw from the predictive (2) and Φ is the standard normal cumulative distribution function. The plotted per-site value is the *centred* PIT of the final observation, $c_s = 2u_{T_s} - 1 \in [-1, 1]$.

Why this alternative. It has the same intent and the same $[-1, 1]$ range as δ_s , but avoids its caveats:

- **Out of sample:** u_t uses the forecast made before y_t is assimilated.
- **Includes observation error:** the predictive adds R_t .
- **Uses the whole distribution:** the full mixture predictive, including the disturbance components and the integration over process error, not a symmetric interval.

If observations really come from the predictive, u_t is uniform on $[0, 1]$ whatever the predictive's shape. So c_s is uniform on $[-1, 1]$, and exactly 95% of sites should have $|c_s| \leq 0.95$.

Panels.

- Page 2 pools the final-observation PIT over every run, i.e. every (experiment, site) pair. The left panel is the histogram of u_{T_s} over all runs, with the share of runs with $|c_s| \leq 0.95$ and the range expected from sampling noise in the title. The right panel is the coverage-error curve of Section 3.3, one line per experiment and a thick line for all runs. (The test-run version of this page, Figure 1, still plots c_s against y_{T_s} per site.)
- Page 3 is the PIT histogram: all u_t , pooled over sites and years, as a density on $[0, 1]$. A calibrated filter gives a flat histogram at height 1 (dashed). The shapes read as:

- U-shape: the predictive is too narrow (overconfident);
 - central hump: it is too wide;
 - mass piled on one side: it is biased. Too many small u_t means the forecast is too high.
- Page 5 summarises each experiment: the share of sites with $|\delta_s| \leq 1$, the median $|\delta_s|$, the share of sites with $|c_s| \leq 0.95$, and the same share over all site-years. The last row pools all experiments.

3.3 Coverage error of the central predictive intervals

The share of runs with $|c_s| \leq 0.95$ checks one interval, the central 95%. The coverage-error curve (page 2, right) runs the same check at every level, and so tests the whole shape of the predictive, not only its tails.

Formula. Write F for the cumulative distribution function of the predictive (2). For a nominal level $p \in (0, 1)$, the *central p -interval* is the middle fraction p of the predictive, $[F^{-1}(\frac{1-p}{2}), F^{-1}(\frac{1+p}{2})]$. The observation falls inside it exactly when

$$\frac{1-p}{2} \leq u \leq \frac{1+p}{2} \iff |2u - 1| \leq p \iff |c| \leq p,$$

so no quantiles need to be computed: the PIT alone decides it. Over a set of n runs with centred PITs $c_1, \dots, c_n$, the empirical coverage and the coverage error at level p are

$$\hat{\kappa}(p) = \frac{1}{n} \sum_{r=1}^n \mathbf{1}\{|c_r| \leq p\}, \quad e(p) = \hat{\kappa}(p) - p.$$

The curve plots $e(p)$ at $p = 0.05, 0.10, \dots, 0.95$ and 0.99 . The 95% check of page 5 is the single point $\hat{\kappa}(0.95)$.

What a calibrated filter gives. If u is uniform on $[0, 1]$, then $\Pr(|c| \leq p) = p$ for every p , so $\hat{\kappa}(p) \approx p$ and $e(p) \approx 0$ across the whole curve. A curve that stays inside the grey band at every level is consistent with a calibrated predictive.

How to read the sign.

- $e(p) > 0$: the observation lands inside the central p -interval *more* often than the fraction p . The predictive is **too wide** at that level (underconfident): its intervals are wider than the forecast errors require.
- $e(p) < 0$: the observation lands inside *less* often than p . The predictive is **too narrow** (overconfident), or centred in the wrong place.
- A curve of one sign at every level is a pure spread problem: negative throughout for a predictive that is too narrow (a U-shaped PIT histogram), positive throughout for one that is too wide (a hump at $u = 0.5$).
- A curve that is negative at small p and positive at larger p means the PITs pile up at a fixed distance from the centre. For example, if most u lie near 0.3 (so $|c| \approx 0.4$), the observation misses the narrow intervals with $p < 0.4$ but falls inside every interval with $p > 0.4$. This is the signature of a **biased** predictive that is also too wide, and it is what the full run shows (Figure 5).

Because $|c|$ throws away the sign of c , the curve cannot say *which* way a bias goes: $u = 0.3$ and $u = 0.7$ count the same. The histogram on the left of the same page shows the direction (mass below 0.5 means the forecast is too high).

Grey bands. Even a calibrated filter gives $e(p) \neq 0$ from sampling noise. If the n runs are independent and calibrated, $n\hat{\kappa}(p) \sim \text{Binomial}(n, p)$, and the band at level p runs between the binomial 2.5% and 97.5% quantiles divided by n , minus p . The light band uses n for one experiment (the smallest experiment, one run per site); the darker band uses n for all runs pooled. Two cautions:

- The bands are pointwise. Over the 20 levels, a calibrated curve can step outside at one or two of them by chance, so look for a systematic departure, not a single point.
- The pooled band assumes the pooled runs are independent. They are not: the eight experiments filter the same sites and observations, so their PITs are correlated, and the true sampling spread of the pooled curve is wider than the dark band. The per-experiment band does not have this problem, because each experiment has one run per site.

4 Plot 2: probability of detection vs. absolute disturbance size

Detection. Let t_{04} be the timestep for 2004 ($t_{04} = 15$). The posterior probability that site s was disturbed in 2004 is

$$P_s = \Pr(k_{t_{04}} \geq 1 \mid y_{1:t_{04}}) = \sum_g \pi_{t_{04}}(g) \sum_j \sum_{k \geq 1} w_{t_{04}}^A(g, j, k),$$

stored as `hist$PDist` at `hist$year == 2004`. It is a filtering probability: it uses the observations up to and including 2004.

Size. The x-axis is the absolute (not proportional) change in observed AGB from 2003 to 2004:

$$\Delta_s = y_{s,2003} - y_{s,2004} \quad (\text{kg C m}^{-2}),$$

which is positive for a loss. The sign is kept so that gains are visible. Points are coloured by the product's 2004 label for the site: none, other, cut, fire or biotic.

Fitted curve (20 or more sites only). When an experiment has at least 20 sites, the script fits a beta regression. This is a regression for responses in $(0, 1)$ that are beta-distributed with mean μ_s and precision ϕ :

$$\tilde{P}_s \sim \text{Beta}(\mu_s \phi, (1 - \mu_s) \phi), \quad \text{logit } \mu_s = \log \frac{\mu_s}{1 - \mu_s} = a + b \Delta_s.$$

The response is shrunk into the open interval so the beta likelihood is finite: $\tilde{P}_s = (P_s(n-1)+0.5)/n$, with n the number of sites. The panel shows the fitted mean $\mu(\Delta)$ and the band between the fitted beta distribution's 2.5% and 97.5% quantiles. The summary table (page 5) lists $\Delta_{50} = -a/b$, the loss detected with probability one half. With 5 sites no curve is fitted.

Reading it. With the product, a flag raises that year's $\beta_{k,t}$ to the user's accuracy. So a flagged site can reach a high P_s even with a small Δ_s . Without the product, P_s is high only when Δ_s is large compared with the spread of the undisturbed forecast.

Panels. Page 4 of `exps_summary.pdf`, one panel per experiment.

5 Experiment grid and test run

The `run-grid` chunk of `OR_joint_update2.Rmd` runs all $2 \times 2 \times 2 = 8$ experiments on the sites in `grid.sites`:

factor	levels
magnitude of R_t	GEDI, LandTrendr
form of R_t	heteroskedastic (het), constant (con)
disturbance product	used (TRUE), not used (FALSE)

(m_R, b_R) : GEDI het (0.0133, 0.0976), GEDI con (0, 0.885), LandTrendr het (0.0133, 3.013), LandTrendr con (0, 3.800).

All runs use the Gamma τ prior above (`grid.tauPrior = "gamma"`). The test run used one randomly chosen site (seed 2004) for each of the five 2004 product labels. Each (experiment, site) result is saved to `experiments/testRun` as

```
site_<idx>_distprod_<TF>_<source>_<model>_tau<prior>.rds
```

and `OR_joint_update_analysis.Rmd` reads that folder (`indir`) and writes `exps_summary.pdf` there. For a full run, set `grid.sites <- seq_len(nrow(dextract))`.

Test-run results

site	2004 product label	Δ_s (kg C m ⁻²)	y_{2017} (kg C m ⁻²)
11	none	-0.15	8.25
118	other	0.30	2.55
212	cut	17.85	10.80
319	fire	0.80	2.60
402	biotic	0.10	1.90

With five sites these plots only show what each panel looks like and give a first sanity check. They are not a calibration result.

- **Filter divergence.**

- Every site is inside $[-1, 1]$ in all eight experiments. The median $|\delta_s|$ is 0.20–0.52.
- It is smallest with GEDI heteroskedastic R (0.20 and 0.23), whose small R_t lets the analysis follow the observation most closely.
- The sign follows the site: the high-biomass sites 11 and 212 have $\delta_s > 0$, and the three low-biomass sites have $\delta_s < 0$.

- **Final-observation PIT.** Every $|c_s| \leq 0.95$ as well ($|c_s| = 0.20$ –0.84).

- **PIT histogram** (140 site-years per experiment). This is where the PIT shows what the divergence does not.

- The histograms are not flat. Most mass lies between 0.2 and 0.4.

- Without the product, 82–100% of the observations at sites 11, 118, 319 and 402 fall below the forecast median (GEDI heteroskedastic R). So the one-step forecast is biased high: VSEM grows these sites faster than observed, and the analysis pulls the state back each year.
- The share of u_t in $[0.25, 0.75]$ is 0.65–0.78 with LandTrendr-magnitude R , against 0.5 expected, so that predictive is also too wide. With GEDI heteroskedastic R it is 0.37–0.49.
- With the product, a second peak appears near $u = 0.8$. It comes from flagged years with little observed loss: the product’s flag puts predictive mass on low biomass, so an unchanged observation lands in the upper tail (e.g. site 118 in 2004: $u = 0.82$ with the product, 0.32 without).

- **Detection.**

- Site 212 has $P_s = 1.000$ in all eight experiments: its loss of 17.85 is detected from the biomass data alone.
- The other sites lose at most 0.8. Without the product they have $P_s \leq 0.061$.
- With the product, the flagged sites reach 0.36–0.83, and the unflagged site 11 stays at 0.015–0.035.

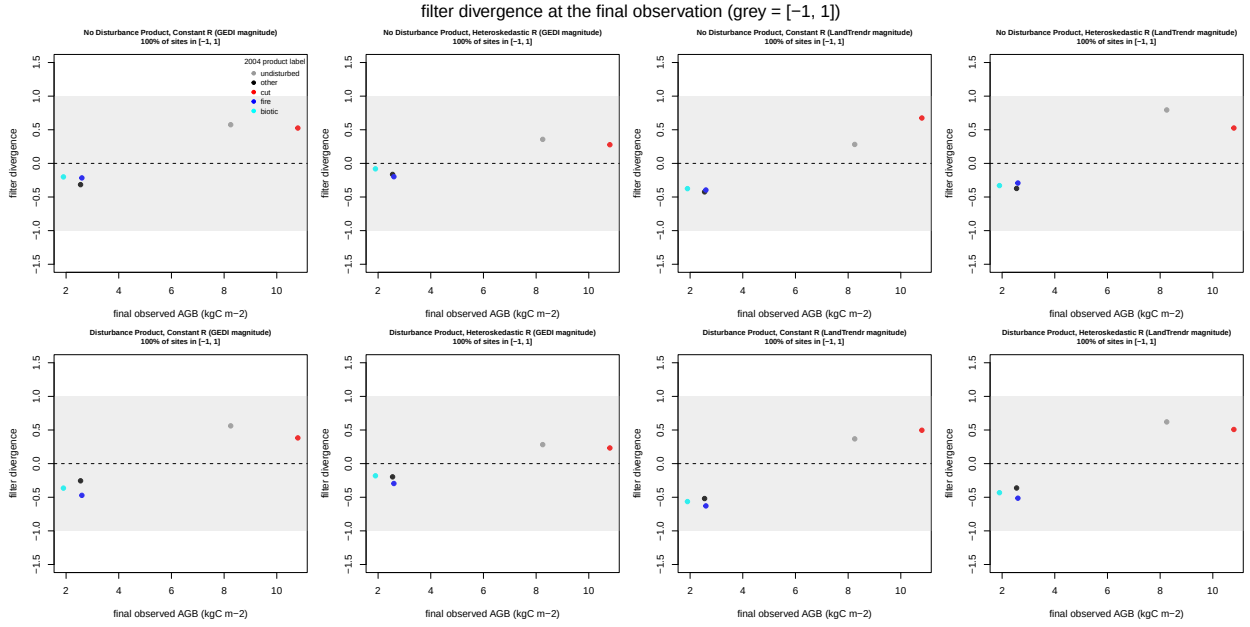


Figure 1: Filter divergence δ_s at the final observation, one point per site and one panel per experiment (test run).

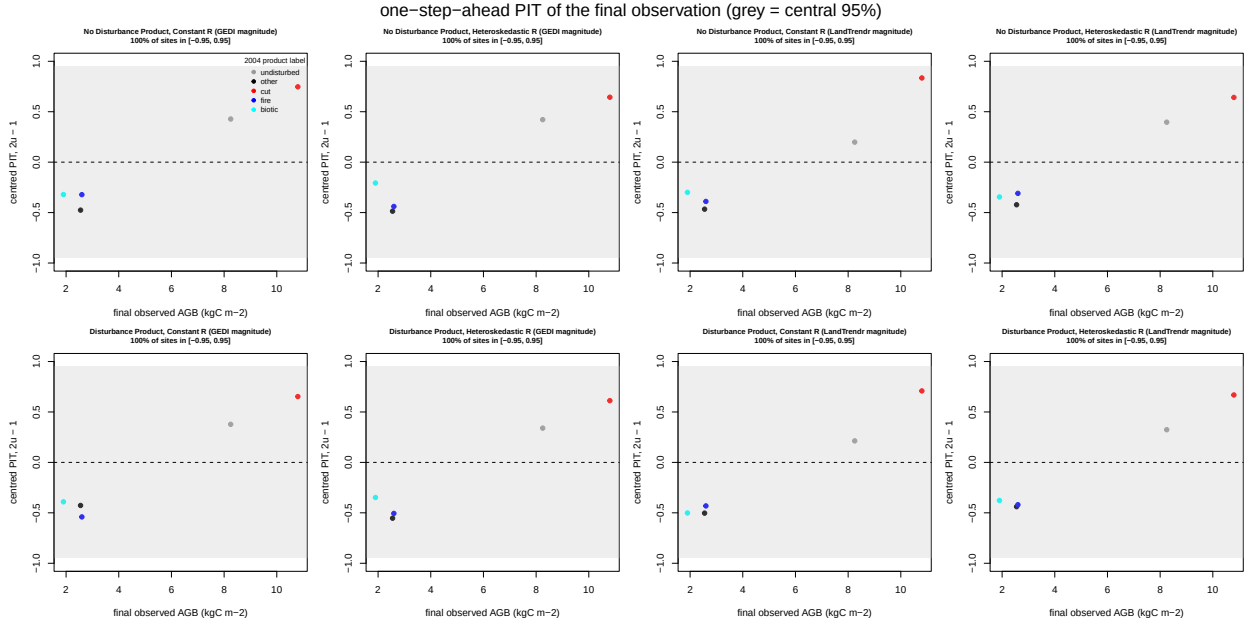


Figure 2: Centred PIT c_s of the final observation, one point per site (test run).

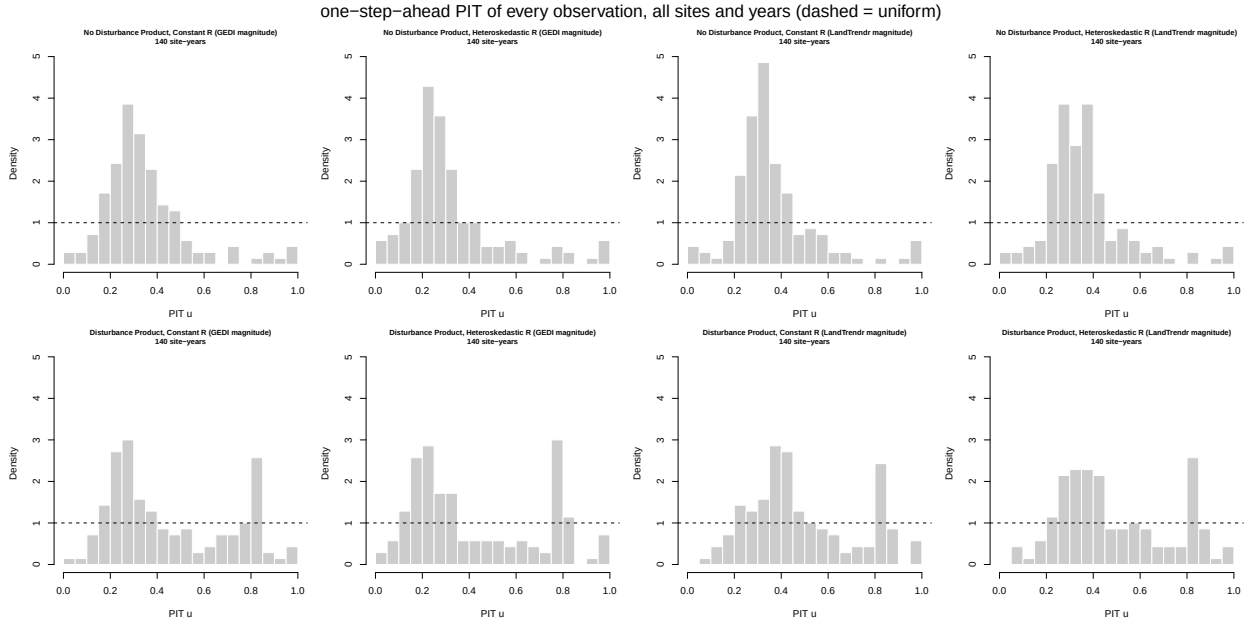


Figure 3: PIT histograms over all sites and years (test run).

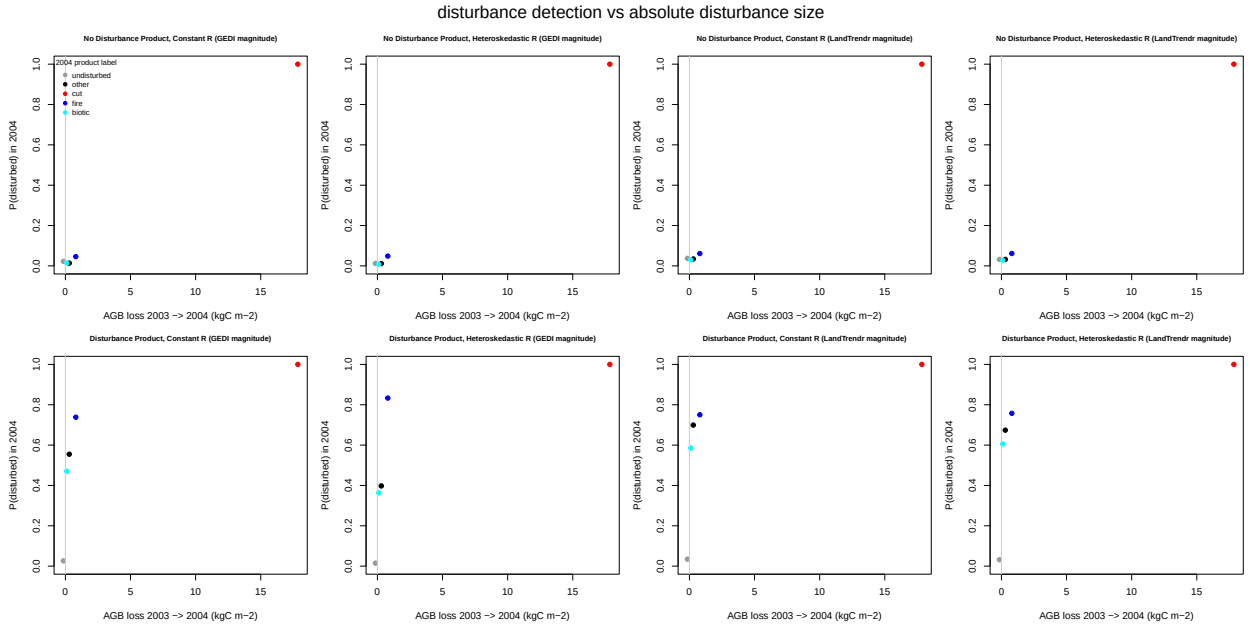


Figure 4: Posterior probability of disturbance in 2004 against the observed AGB loss from 2003 to 2004 (test run).

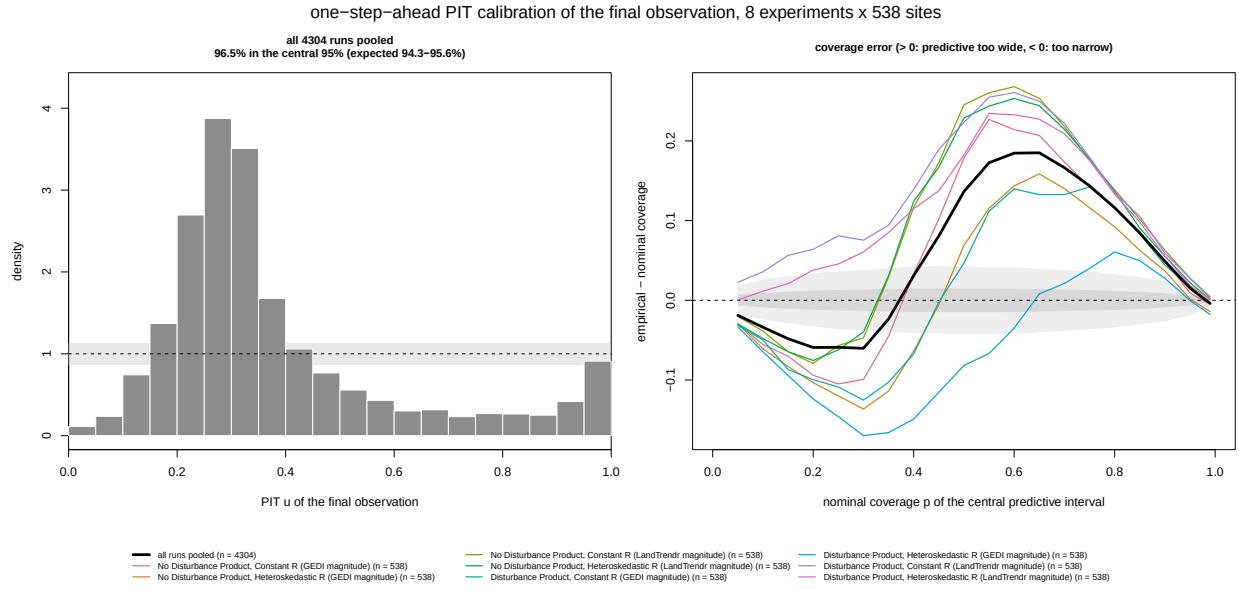


Figure 5: Page 2 from the full run (538 sites $\times$ 8 experiments, 4,304 runs). Left: histogram of the final-observation PIT over all runs. Right: coverage error $e(p)$ of the central predictive intervals (Section 3.3), one thin line per experiment and a thick line for all runs pooled. 96.5% of runs fall in the central 95% (slightly above the 94.3–95.6% expected), yet the pooled curve is negative for $p \leq 0.35$ (down to -0.06) and reaches $+0.19$ at $p = 0.6$ – 0.65 : the PITs bunch near $u \approx 0.3$, so the forecast is biased high and too wide.